# Aspect-Based Sentiment Analysis for Automated Product Review Analysis Using Natural Language Processing

Project presentation

# The problem

*The screen is gorgeous and the keyboard feels great, but the battery life is terrible.*

- Traditional sentiment analysis: one label for the whole review
- What a seller needs: screen positive, keyboard positive, battery life negative
- Reading every review by hand does not scale

# Objectives

1. Extract product aspects with a learnt model, with no hard-coded aspect list
2. Classify the sentiment towards each aspect
3. Measure both on a standard benchmark and against a non-neural baseline
4. Deliver a working web application

# Dataset: SemEval-2014 Task 4

- Laptop and restaurant review sentences, annotated by people
- Aspect terms with character offsets and polarity
- Official gold test set used unchanged; duplicates and train/test overlaps removed

| Domain | Training | Validation | Test |
|---|---|---|---|
| laptops | 2723 | 303 | 800 |
| restaurants | 2726 | 303 | 800 |

# Method: two stages

```
review -> clean, split, tokenise
       -> BERT token classifier     (BIO tags: which words are aspects?)
       -> BERT sentence-pair model  ([CLS] sentence [SEP] aspect [SEP] -> sentiment)
       -> aspect-sentiment map + overall sentiment
```

- Both start from bert-base-uncased, fine-tuned end to end
- A third model gives the traditional single label, for comparison

# Why BIO tags

| The | battery | life | is | great | but | the | screen | is | dim |
|---|---|---|---|---|---|---|---|---|---|
| O | B-ASPECT | I-ASPECT | O | O | O | O | B-ASPECT | O | O |

- Multi-word aspects are one unit
- Adjacent aspects stay separate
- No stop-word removal: BERT needs "not" and "but"

# System

```
React + TypeScript  ->  FastAPI  ->  ABSA pipeline (3 BERT models)
                                 ->  SQLite (review history)
```

- Analyzer, batch CSV analysis, history, dashboard
- Every metric and document is regenerated by script
- Automated tests for backend and frontend

# Results on the test sets

| Metric | Both domains | Laptops | Restaurants |
|---|---|---|---|
| Aspect extraction F1 | 0.846 | 0.802 | 0.871 |
| Aspect sentiment accuracy | 0.815 | 0.774 | 0.838 |
| Aspect sentiment macro-F1 | 0.746 | 0.732 | 0.753 |
| End-to-end F1 | 0.698 | 0.631 | 0.735 |

Exact span match; end-to-end requires aspect and sentiment both right.

# What BERT adds

Baseline: noun phrases as aspects, opinion-word lexicon for sentiment. Same data, same metrics.

| Metric | Baseline | BERT |
|---|---|---|
| Aspect extraction F1 | 0.376 | 0.846 |
| Aspect sentiment accuracy | 0.663 | 0.815 |
| Aspect sentiment macro-F1 | 0.616 | 0.746 |
| End-to-end F1 | 0.253 | 0.698 |

# Transfer to an unseen domain

Restaurant test set: an aspect extractor trained on laptops only against the one trained on both domains.

| Metric | Trained on laptops only | Trained on both |
|---|---|---|
| Aspect extraction precision | 0.892 | 0.848 |
| Aspect extraction recall | 0.234 | 0.897 |
| Aspect extraction F1 | 0.370 | 0.871 |

# Where it still fails

- Aspects in sentences with one sentiment: accuracy 0.851
- Aspects in sentences that mix sentiments: accuracy 0.662
- Negative class F1: 0.749
- Neutral class F1: 0.581
- Positive class F1: 0.906
- Only explicit aspects are found

# Limitations

- Training data: two domains, from 2014
- One training run, one seed, no confidence intervals
- Overall-sentiment labels are derived, not annotated
- Confidence values are not calibrated

# Demo

1. Analyze a laptop review and a restaurant review
2. A review with no aspect
3. Batch: upload a CSV, export the results
4. History and dashboard

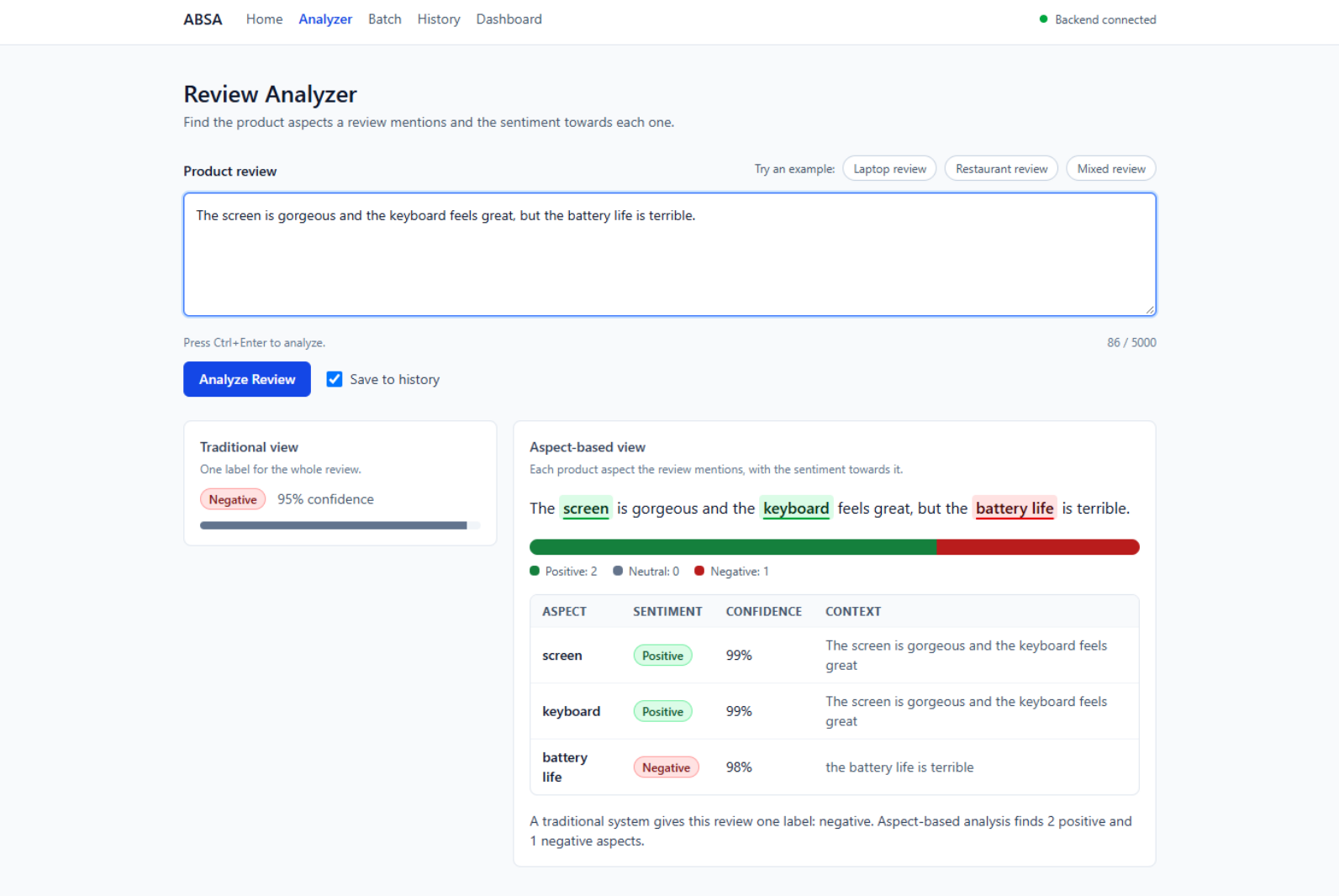
Script: docs/demo.md

# Conclusion and future work

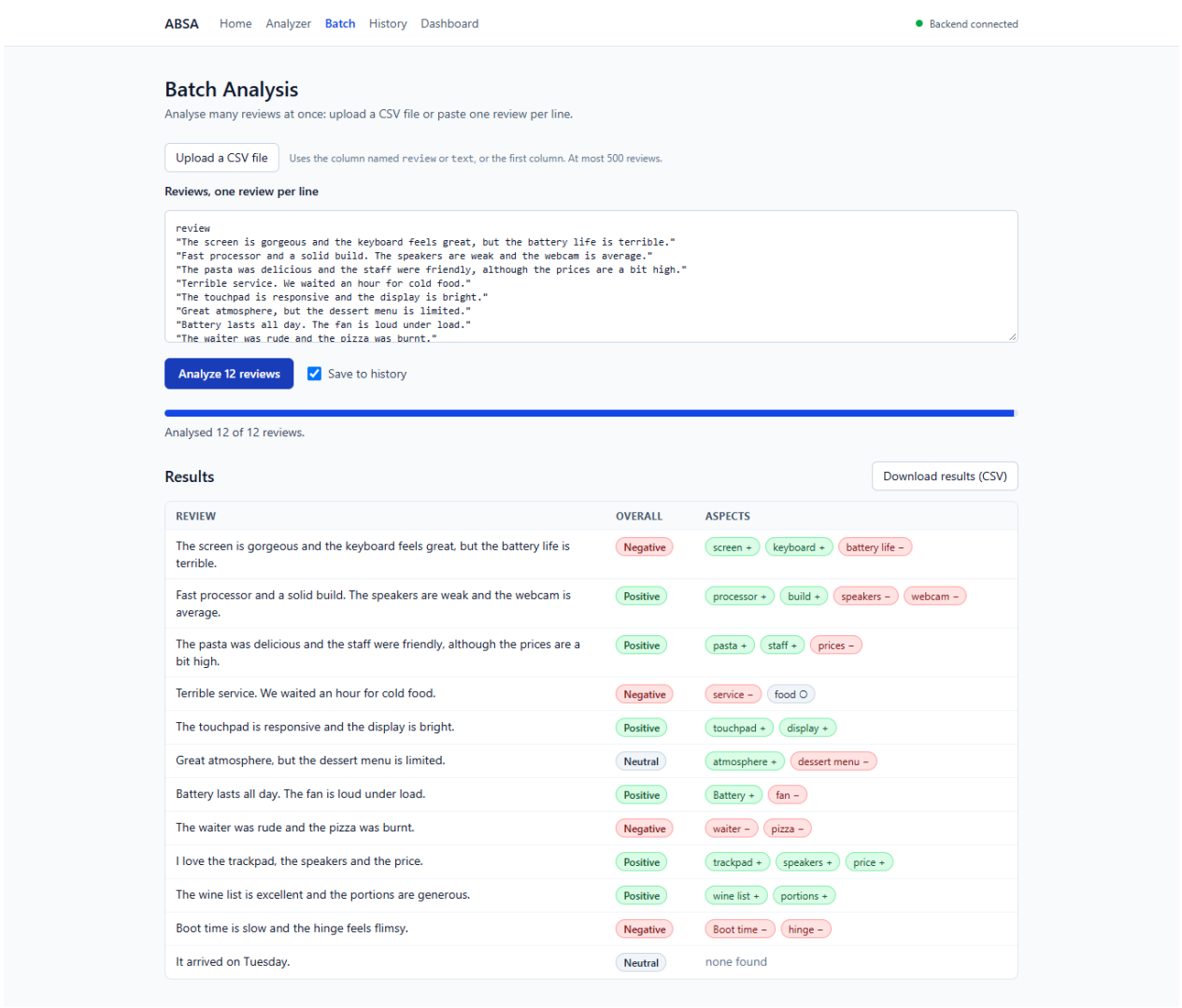
- A complete, tested ABSA system with reproducible evaluation
- Next: more data with contrasting aspects, aspect position in the input, several seeds, implicit aspects, more product categories

Thank you. Questions?

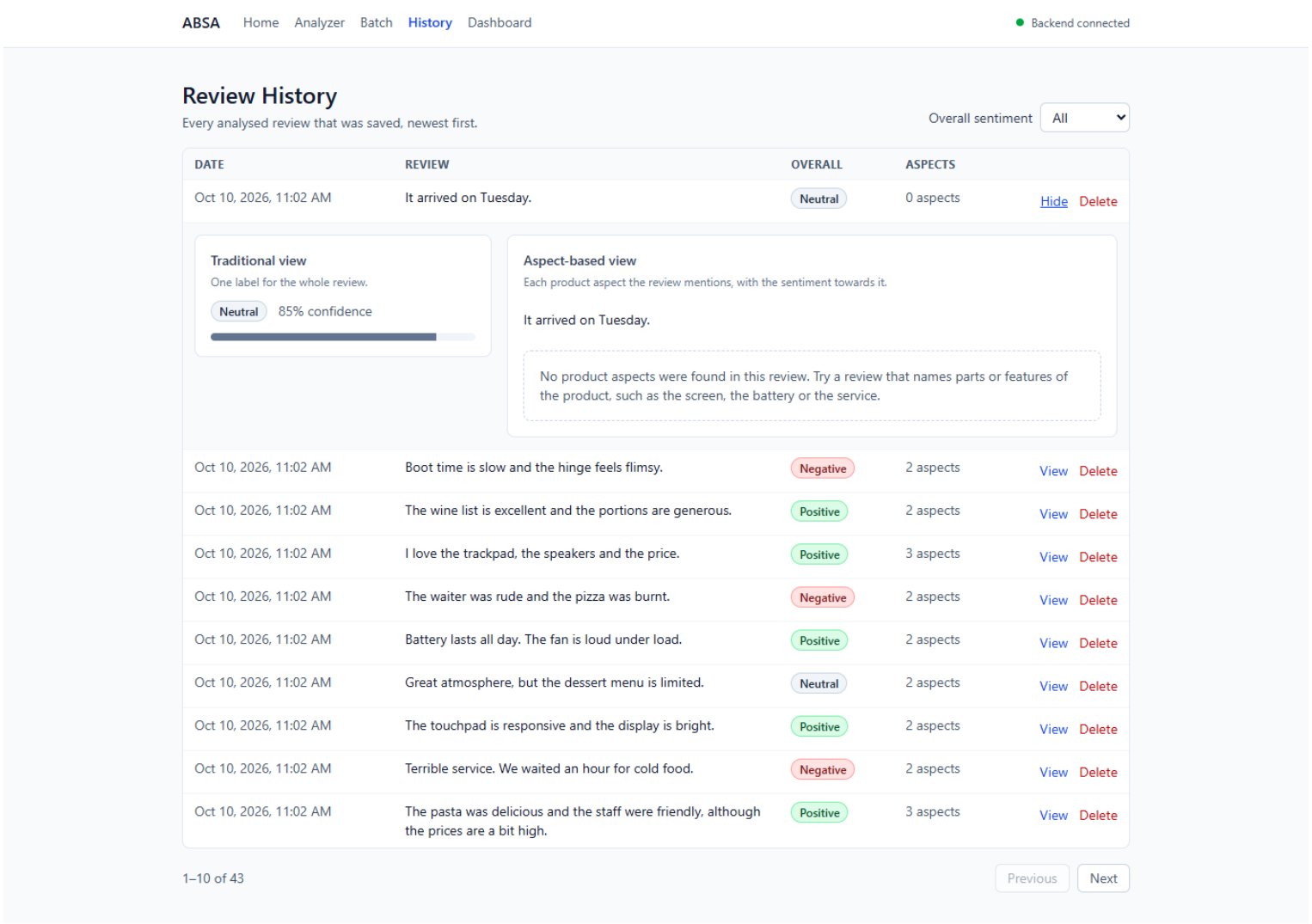
# Application: analyzing one review



# Application: batch analysis of a CSV file



# Application: review history



# Application: dashboard and model performance

